



**Politecnico
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Master's Degree in Mathematical Engineering

Material for Thesis

b– Data Pre-processing Pipeline
Intra and Inter Analysis

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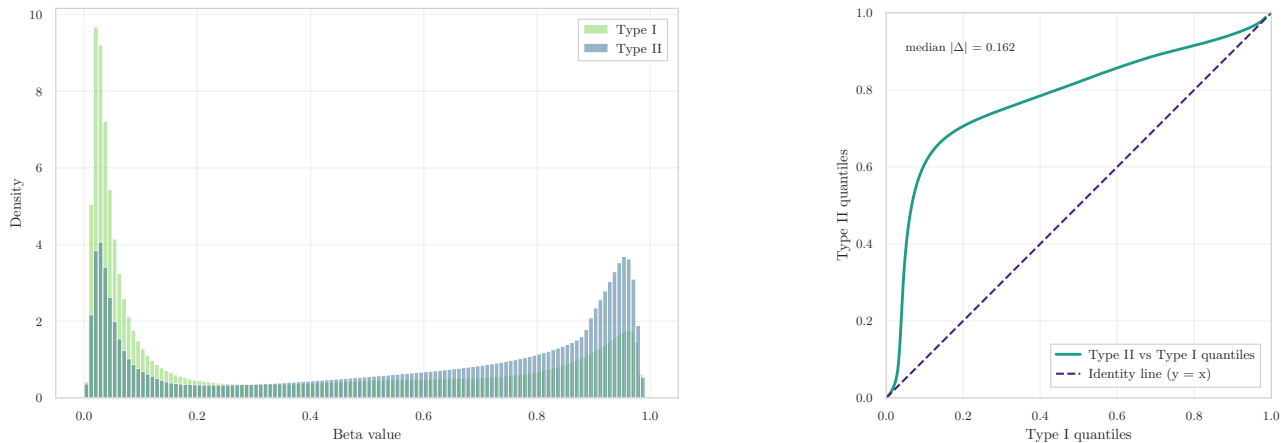
Abstract

MANCA L'ABSTRACT.

1 Intra-Dataset Pre-processing

Data pre-processing represents a foundational stage in DNA methylation analysis, particularly in cancer epigenomics, where technical artefacts, probe-design biases, and unwanted sources of variation can obscure or distort genuine biological signals. Before performing differential analyses, feature selection, or machine-learning models, it is essential to ensure that the methylation matrices used for statistical modelling are accurate, harmonised across platforms, and free from unreliable probes or batch-associated effects.

In the context of breast cancer, where early epigenetic alterations may be subtle and highly localized, careful pre-processing is crucial to avoid confounding Normal-Adjacent differences with technical noise. Several well-established



(a) β -value distribution by Infinium probe design (Type I vs. Type II). The near-overlapping shapes indicate effective correction of probe-type bias. (b) Q-Q plot comparing quantiles of Type II vs. Type I probes. The alignment to the diagonal ($y = x$) confirms balanced signal distributions after normalization.

Figure 1: Diagnostic evaluation of Infinium Type I/II probe bias correction in the GSE69914 dataset. The consistent β -value and quantile distributions demonstrate the effectiveness of the BMIQ normalization previously applied.

sources of unreliability—cross-reactive probes, SNP-affected loci, non-CpG targets, Infinium Type I/II probe-design bias, and batch effects—must be addressed through a systematic and reproducible workflow.

The primary goal of this section is therefore to construct, for each dataset, a high-quality and biologically coherent methylation matrix suitable for downstream analyses. To this end, all datasets undergo the same structured pipeline: initial alignment and quality checks; removal of unreliable CpGs through literature-based technical filtering; conversion from β -values to M-values for statistical modelling; correction of batch effects; and identification of invariant CpGs that carry no discriminative information along the Normal-Adjacent axis.

These steps ensure methodological consistency across the three cohorts (GSE69914 [1.1], GSE225845 [1.2], GSE287331 [1.3]) and provide a unified, denoised, and biologically meaningful set of features to be used in the subsequent analyses of early epigenetic alterations.

1.1 Dataset GSE69914

Step 0 – Initial Pre-processing The initial pre-processing stage verifies the structural integrity of the dataset. The β -value matrix and its corresponding phenotype table were imported from their Parquet representations. Both tables contained ✓ 407 samples, and the β -matrix included ✓ 485,512 CpGs, confirming full dataset availability at load time.

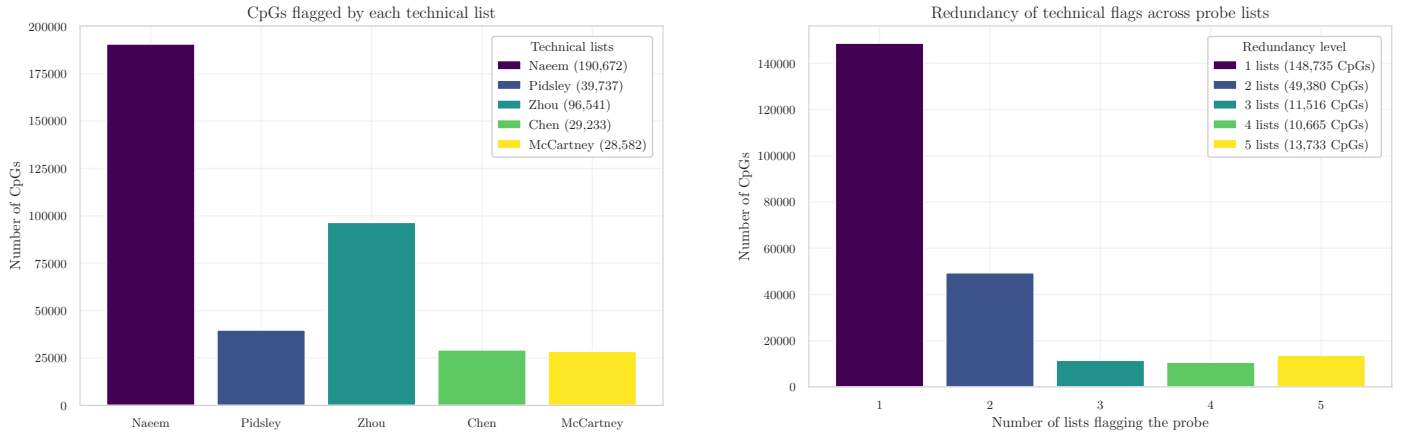
Sample identifiers in the phenotype table were inspected to ensure uniqueness and correct metadata structure. No duplicated `id.tissue` entries were detected ✓ (0 duplicates), and all required fields (`id.tissue`, `label`) were present ✓ (0 missing fields), indicating that the phenotype annotation is complete and coherent. Alignment between the phenotype table and the β -matrix was assessed by intersecting sample IDs. All samples were shared across matrices, yielding ✓ perfect correspondence (407/407 matched).

A probe-level and sample-level missingness check was then performed to verify the completeness of the methylation matrix. The entire dataset was found to be fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. This confirms that no early NaN filtering or imputation is required for this dataset.

Step 1 – Infinium Type I/II Bias Diagnostics Raw methylation intensity data (*IDAT* files) were processed by the original authors using the `minfi` package (v1.8.9) and BMIQ normalization (v1.4), as reported in [1]. This preprocessing pipeline includes the **bias correction between Infinium Type I and Type II probes**, ensuring that β -value distributions from both probe designs are comparable prior to downstream analysis.

I independently verified that this probe-type bias correction had been successfully applied. To this aim, the β -value distributions were stratified by probe design (Type I vs. Type II) using the official Illumina 450K manifest obtained from the Bioconductor package [2]. Following the diagnostic framework proposed by Teschendorff et al. [3], two diagnostic plots were generated:

1. The β -value density distributions of Type I and Type II probes (Fig. 1a) show extensive overlap, with no evident design-driven bias in the normalized data;



(a) CpGs flagged by each technical filtering resource (Naeem, Pidsley, Zhou MASK_*, Chen, McCartney).

(b) Redundancy distribution of probes flagged by 1–5 lists.

Figure 2: Summary of technical filtering for the GSE69914 methylation dataset. **(a)** Intersection between the dataset and each external probe-filtering resource, showing the number of CpGs flagged by Naeem, Pidsley, Zhou MASK_*, Chen, and McCartney lists. **(b)** Redundancy analysis indicating how many probes were simultaneously flagged by 1 to 5 independent resources, quantifying the concordance among technical annotations.

2. The Q–Q plot comparing Type II versus Type I quantiles (Fig. 1b) exhibits a near-diagonal alignment, confirming effective bias correction.

The observed post-normalization pattern closely reproduces the characteristic outcome described by Teschendorff et al. [3] and Wang et al. [4], who demonstrated that BMIQ substantially reduces Infinium Type II probe bias, yielding overlapping β -value distributions and near-diagonal Q–Q relationships. ✓ Hence, this verification confirms that the dataset underwent proper BMIQ normalization.

Step 2 – Technical Filtering Technical filtering aims to remove unreliable or biologically confounded probes prior to statistical modelling. This stage reduces noise, improves downstream reproducibility, and ensures that only high-confidence CpG loci are retained for analysis (Fig. 2).

Exclusion of technical probe sets A collection of curated probe resources was used to exclude loci known to exhibit technical artefacts (see Appendix A for details). These include probes affected by common genetic variants at the CpG or single-base extension site [5], [6], probes with demonstrated multi-mapping or off-target hybridisation [7], [8], and probes flagged by the MASK_* reliability annotations [5]. Additional quality control criteria follow the hierarchical filtering strategy of Naeem et al. [9], which targets multi-mapping loci, repeat structures, and disruptive SNPs.

The overlaps between these resources and the β -matrix were substantial (Fig. 2a). A total of ✓ 190,672 probes were flagged by Naeem et al., ✓ 39,737 by Pidsley et al., ✓ 29,233 by Chen et al., ✓ 96,541 by the Zhou MASK_* fields, and ✓ 28,582 by McCartney et al. These lists are known to partially overlap—particularly those focused on multi-mapping and SNP-associated unreliability—and their union resulted in the removal of ✓ 225,426 unique CpGs. After this operation, ✓ 260,086 CpGs remained.

Annotation-based filtering (Illumina manifest) To ensure consistency with the manufacturer’s official annotation, the HumanMethylation450 v1.2 manifest [10] was used to validate probe identity and genomic mapping. Through this annotation, probes lacking valid chromosome information, those with undefined genomic coordinates, and those targeting non-CpG contexts (IDs beginning with “ch”) were identified and removed. This step aligns the working feature set with Illumina’s reference genome definition and follows the recommendations of Zhou et al. [5] and Pidsley et al. [6]. A total of ✓ 875 non-CpG probes were excluded, reducing the feature space to ✓ 259,211 CpGs.

Removal of sex-chromosome probes (X/Y) To avoid sex-related confounding effects and to maintain cross-dataset consistency, all probes located on chromosomes X and Y were removed based on manifest annotations. This choice is consistent with established preprocessing workflows for Illumina 450K data, where probes on sex chromosomes are excluded when male and female samples are mixed, as recommended by Maksimovic et al. [11]. Moreover, sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and joint normalization of autosomal and sex-chromosome probes has been shown to introduce artificial sex-related bias into thousands of autosomal CpGs

[12]. Removing X/Y probes therefore provides a conservative and robust strategy when sex-specific methylation is not under investigation. This filtering step eliminated ✓ 7,728 CpGs, producing a final autosomal working set of ✓ 251,483 CpGs.

Technical filtering summary Overall, the initial set of ✓ 485,512 CpGs was reduced to ✓ 251,483 autosomal CpGs after applying all technical lists, manifest-based validation, and sex-chromosome exclusion. In total, ✓ 234,029 unique CpGs were removed across all criteria. A per-probe summary file [removed.cpgs.all.filters.summary.gse69914.csv](#) records, for each excluded probe, which resources contributed to its removal, ensuring full reproducibility of the filtering pipeline.

Step 3 – Invariance Filtering on Normal+Adjacent

Selection of the NA Subset

Variability Metrics (r_β , variance)

Threshold Selection

Creation of Final CpG Mask

Step 4 – β -to- M Conversion

Definition of β_{clean}

Logit Transformation to M -values

Use of β and M -values in Downstream Analyses

Step 5 – Batch Effect Correction

Pre-batch Diagnostics

Definition of the Batch Model

ComBat Correction

Post-batch Verification

Step 6 – Final Outputs

Final Matrices ($\beta_{\text{final_preFS}}$, $M_{\text{final_preFS}}$)

CpG Tracking

Mini-report Summary (optional)

1.2 Dataset GSE225845

Step 0 — Initial Pre-processing

Ingestion and Alignment

Coarse NaN Assessment

Infinium Type I/II Bias Diagnostics

Step 1 — Technical Filtering

Cross-reactive Probes

EPIC Unreliable Probes

SNP-affected Probes

Non-CpG Probes

Sex-chromosome Probes

Technical Filtering Summary

Step 2 — β -to- M Conversion

Definition of β_{clean}

Logit Transformation

Usage of β and M-values

Step 3 — Batch Effect Correction

Pre-batch Diagnostics

Batch Model Definition

ComBat Application

Post-batch Verification

Step 4 — Invariance Filtering on Normal+Adjacent

NA Subset Selection

Variability Metrics

Threshold Selection

Final CpG Mask

Step 5 — Final Outputs

Final Matrices

CpG Tracking

Mini-report Summary (optional)

1.3 Dataset GSE287331

Step 0 — Initial Pre-processing

Ingestion and Alignment

Coarse NaN Assessment

Infinium Type I/II Bias Diagnostics

Step 1 — Technical Filtering

Cross-reactive Probes

EPIC Unreliable Probes

SNP-affected Probes

Non-CpG Probes

Sex-chromosome Probes

Technical Filtering Summary

Step 2 — β -to- M Conversion

Definition of β_{clean}

Logit Transformation

Usage of β and M-values

Step 3 — Batch Effect Correction

Pre-batch Diagnostics

Batch Model

ComBat Application

Post-batch Verification

Step 4 — Invariance Filtering on Normal+Adjacent

NA Subset Selection

Variability Metrics

Threshold Selection

Creation of Final Mask

Step 5 — Final Outputs

Final Matrices

CpG Tracking

Mini-report Summary (optional)

2 Inter-Dataset Analysis

3 Data Validation and Integrity Check

Missing Value Analysis. Next, I performed a comprehensive check for missing values (NaN), as these can severely impact model performance and must be addressed before training.

- No missing entries (NaN) were detected across any CpG probe. ✓ Total NaN count: 0 — Overall missing rate: 0%.
- The methylation matrix is therefore **complete**, requiring **no filtering or imputation** procedures. ✓ Dataset confirmed fully complete.
- For future datasets:
 - If the overall missing rate is $< 1\%$, imputation may be considered as an optional step.
 - If probe missingness exceeds 5% or sample missingness exceeds 10%, the affected entities should be discarded, following standard preprocessing practices [5].

This validation confirms the dataset is structurally sound, numerically consistent, and complete, enabling unbiased downstream variance modeling, differential methylation testing, and batch correction without any further cleaning.

4 Comparison of Beta-value and M-value Quantifications

Illumina Infinium reports two background-corrected signals per CpG, methylated $y^{(M)}$ and unmethylated $y^{(U)}$. The **Beta-value** is the proportion of methylated intensity,

$$\beta_i = \frac{\max(y_i^{(M)}, 0)}{\max(y_i^{(M)}, 0) + \max(y_i^{(U)}, 0) + \alpha}, \quad 0 \leq \beta_i \leq 1,$$

while the **M-value** is the log-ratio,

$$M_i = \log_2 \left(\frac{\max(y_i^{(M)}, 0) + \alpha}{\max(y_i^{(U)}, 0) + \alpha} \right) = \log_2 \left(\frac{\beta_i}{1 - \beta_i} \right).$$

Rationale. As shown by Du *et al.* [13], β is *bounded* and *heteroscedastic* (its variance depends on the mean, especially near 0 and 1), whereas M is approximately *homoscedastic* and therefore more suitable for linear modeling and t -tests. However, β retains higher interpretability since it directly reflects the methylation fraction. Consequently, M -values are adopted for inferential analyses, while β -values are preferred for reporting and visualization.

Implementation in this dataset. The GSE69914 matrix provides β only; therefore, M was derived as

$$M = \log_2 \left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon} \right),$$

using a small $\varepsilon = 10^{-6}$ to ensure numerical stability for extreme β values. ✓ The working matrix was converted to M -values for all subsequent statistical analyses.

Distributional comparison. Figure 3 illustrates the distributional differences between β and M quantifications across Normal and Normal-adjacent tissues. As evident in the upper panels, β exhibits a bounded, U-shaped distribution, while M yields approximately symmetric profiles. The lower panels confirm the pronounced mean-dependent variance of β and the near-homoscedastic behavior of M , ✓ in agreement with Du *et al.* [13].

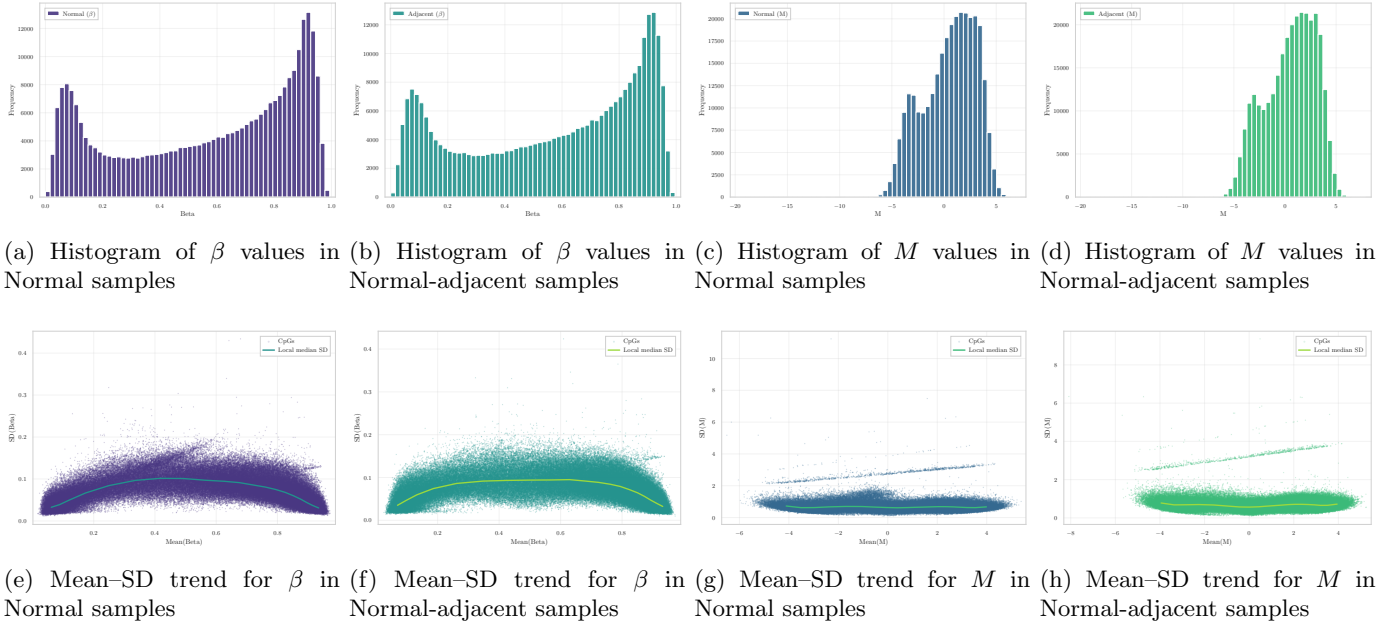


Figure 3: Distributional properties of Normal vs Normal-adjacent tissue methylation. Top row: β -value (bounded fractional) and M -value (log-ratio) histograms illustrate the characteristic U-shaped β distributions and the more symmetric M profiles. Bottom row: Mean-SD relationships across CpGs reveal pronounced mean-dependent heteroscedasticity in β , which is substantially reduced after logit transformation to M , confirming the homoscedasticity advantage described by Du *et al.* [13]. All plots were computed on the current working set of 156,738 CpGs remaining after the filtering steps applied so far.

5 Batch-Effect Correction

Remove inter-array technical variation using **ComBat** (Johnson *et al.*, 2007) or its methylation-specific extension **ComBat-met** (Wang *et al.*, 2025).

If batch effects are confounded with biological groups, include the batch variable as a covariate in linear modeling (e.g., *limma*).

A Filtering lists

This appendix details the major technical filtering lists and annotations applied in Illumina 450K and EPIC methylation arrays to remove unreliable or ambiguous CpG probes.

Table 1: Technical categories covered by each filtering resource.

Category	Naeem A.1	Chen A.2	Pidsley A.3	Zhou A.4	McCartney A.5
Cross-hybridisation / multi-mapping	Yes (Steps 1–2)	Yes	Yes	MASK_mapping	Table 2+3
SNP at CpG / extension base	Yes (Step 5)	–	Yes	MASK_snp5, MASK_extBase	–
Nearby SNP tolerated	Conditional (Step 6)	–	–	–	–
INDEL / structural variant	Yes (Step 3)	–	–	–	–
Non-CpG probes	–	–	Yes	MASK_nonCG	Table 3

A.1 Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the humanmethylation450 array

Naeem *et al.* [9] proposed a structured filtering framework to reduce false discoveries by sequentially discarding probes based on hybridisation specificity and genomic integrity (Figure 4). The main exclusion steps are:

- Multi-mapping or cross-hybridising probes $\Rightarrow$ *discard*.
- Probes overlapping repetitive elements (LINE/SINE/ALU) $\Rightarrow$ *discard*.
- Probes targeting regions with INDELs $\Rightarrow$ *discard*.
- Probes overlapping SNPs:
 - SNP at CpG or extension site $\Rightarrow$ *discard*.
 - SNP near target but not interfering in bisulfite space $\Rightarrow$ *keep*.

The resulting “discard” set therefore removes probes affected by cross-reactivity, structural polymorphisms (INDELs), and SNPs that alter CpG interrogation.

A.2 Discovery of cross-reactive probes and polymorphic cpGs in the illumina infinium humanmethylation450 microarray

Chen *et al.* [7] empirically identified probes in the 450K array that exhibit off-target hybridisation or overlap with common SNPs. For this project, only the **cross-reactive list** is available and used. This list enumerates ~ 29 k multi-mapping probes whose methylation signal cannot be uniquely attributed to one genomic locus.

A.3 Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling

Pidsley *et al.* [6] provide a platform-level assessment of the EPIC array, with explicit *annotated probe lists* that flag probes whose methylation signal may be confounded by design- or genome-related artefacts. The focus is on probe categories and positions where technical bias arises, rather than on a universal, prescriptive blacklist. In particular, they catalogue:

- **Cross-hybridising CpG-targeting probes:** CpG probes showing sequence homology (off-target matches) to additional genomic loci, yielding non-unique hybridisation and potentially inflated or ambiguous β signals.
- **Cross-hybridising non-CpG-targeting probes:** off-target issues among CNG/non-CpG probes; these are typically excluded in CpG-centric analyses due to limited interpretability and higher risk of artefacts.
- **Probes overlapping common genetic variation:** annotation of variants from population data at three critical positions:
 - *At the interrogated CpG* (polymorphic CpG) — directly affects the presence of the CpG dinucleotide and the measured methylation state;

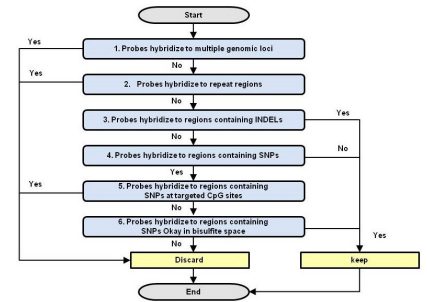


Figure 4: Workflow for determining affected Probes.

- *At the single-base extension (SBE) site* (Type I) — perturbs extension and dye chemistry, biasing intensity ratios;
- *Within the probe body* — can reduce binding affinity or alter hybridisation kinetics, especially for common variants.

A.4 Comprehensive characterization, annotation and innovative use of infinium dna methylation beadchip probes

Zhou *et al.* [5] released a unified probe annotation for both 450K and EPIC arrays with multiple binary mask columns (MASK_*). Each mask flags probes to be removed due to specific technical artifacts.

- MASK.mapping: probes with low or inconsistent mapping quality (non-unique alignment or presence of INDELs);
- MASK.typeINextBaseSwitch: Type-I probes carrying a SNP in the extension base that causes a color-channel switch (CCS probes);
- MASK.extBase: probes whose extension base is inconsistent with the expected color channel or CpG context;
- MASK.sub30.copy: probes with non-unique 30-bp 3' subsequences (potential cross-hybridisation);
- MASK.snp5.common: probes overlapping a SNP within ± 5 bp of the interrogated CpG (even with global MAF $\geq 1\%$);
- MASK.snp5.GMAF1p: probes overlapping SNPs with global MAF $\geq 1\%$;
- MASK.general: recommended composite mask integrating mapping, SNP, and cross-reactivity filters for general use;
- MASK.rmsk15: probes overlapping RepeatMasker regions (not recommended for exclusion in standard workflows).

This resource provides a programmatic and reproducible way to perform fine-grained probe masking.

A.5 Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip

McCartney *et al.* [8] assessed probe design artifacts and published supplementary tables that include:

- Cross-hybridising CpG-targeting probes;
- Cross-hybridising non-CpG-targeting probes.

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